



Spring Framework

Training Assignments

Document Code	25e-BM/HR/HDCV/FSOFT
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RECORD OF CHANGES

No	Effective Date	Change Description	Reason	Reviewer	Approver
1	01/Oct/2018	Add the new labs	Create new	DieuNT1	VinhNV
2	01/Jun/2019	Update template	Fsoft template	DieuNT1	VinhNV

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	CODE:	JSFW.L.A102 (LMS)
	TYPE:	LONG
	LOC:	N/A
	DURATION:	360 MINUTES

Objectives

After finishing the following exercises, trainees will:

- Understand Spring Core (IoC, DI)
- Practice with JDBC Templates to connect database in spring
- Use JPA for object-relational mapping
- Practice with the Spring MVC framework architecture
- Know how to write a Web application with Spring MVC Framework and Spring Data JPA

Business needs

Create a Web application based on HTML, CSS, JavaScript, Ajax, Thymeleaf, Spring framework.

The Course Sharing System that we are going to develop is a single-instructor system (for this assignment).

1. The **Instructor** of the system will be able to perform the following actions:
 - Login and logout
 - Create, update and delete Courses
 - Publish, unpublish and archive Courses
 - Approve and delete Reviews
2. All other users are **Students** (guests) who can perform the following actions:
 - Read Course details and published content
 - Create Reviews (with a 1-5 star rating)
3. Additional Requirements for this system include:

The homepage of the system should display a list of the most recent, published **Courses**.

- If a page contains more than 10 Courses, they should be displayed in pages.
- The system should display a Course together with its approved Reviews.
- The system should be able to list Courses with a specified Category.
- The system should show a cloud of Categories indicating their use frequencies.
- The system should show a list of most recent Reviews.

Working Environments

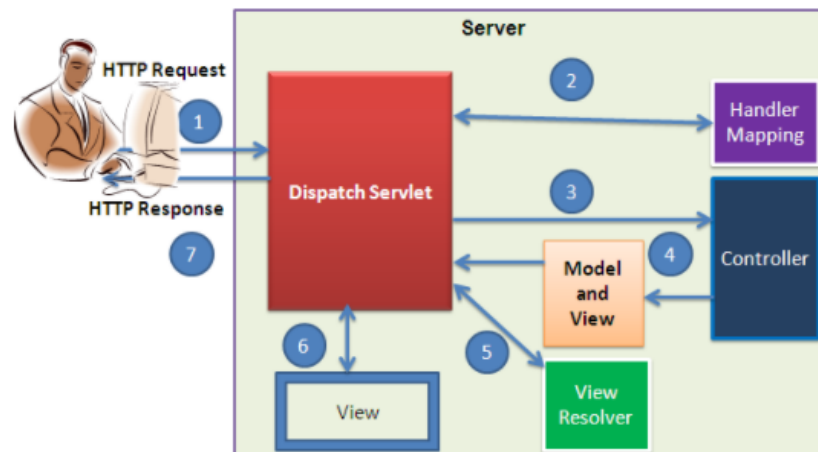
- ✓ JDK 1.8 +
- ✓ Maven 3+
- ✓ Eclipse IDE or STS,
- ✓ MySQL DB Server/MySQL,

- ✓ Apache TomCat 7 or later,
- ✓ Internet connection,

Delivery: Source code, deployment and testing, reviewing evidences packaged in a compress archive.

Product Architecture

Web applications are by nature distributed applications, meaning that they are programs that run on more than one computer and communicate through a network or server. Specifically, web applications are accessed with a web browser and are popular because of the ease of using the browser as a user client.



The product is implemented using MVC Pattern base on Spring Web MVC.

Technologies

The product implements one or more technology:

- HTML & CSS
- JavaScript & Ajax
- Thymeleaf
- Spring Framework: IoC, Bean, SpEL, JdbcTemplate, Spring Data JPA, Interceptor, Validation, Spring Security.

Database Relationship

Based on the analysis of the requirements, we decide to use the following database tables to store the persistent data for our blog application:

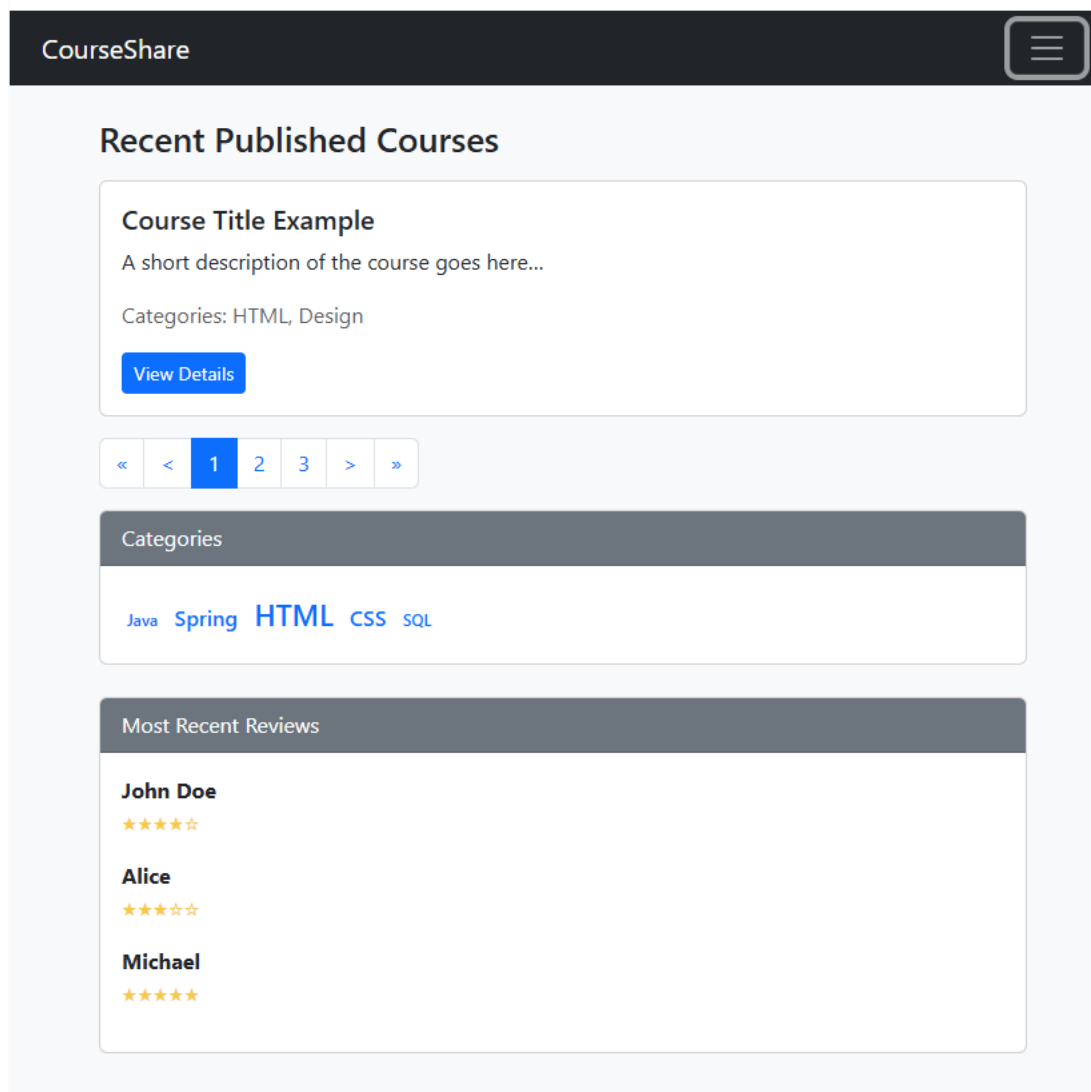
- **tbl_instructor** stores the instructor information, including username and password for login.
- **tbl_course** stores the course information. It mainly consists of:
 - title: required, title of the course.
 - description: required, a short summary of the course.
 - content: required, full details of the course (uses Markdown).
 - status: required, status of the course (e.g., 1=DRAFT, 2=PUBLISHED, 3=ARCHIVED).
 - category: optional, a list of comma-separated words categorizing the course.
- **tbl_review** stores the course review information. Each review is associated with a course:

- author_name: required, the student's name.
- email: required, the student's email.
- rating: required, a score from 1 to 5.
- content: required, the review content in plain text.
- status: required, status of the review (e.g., 1=PENDING, 2=APPROVED).
- **tbl_category** stores course category frequency information (for the category cloud).
 - name: required, the unique category name.
 - frequency: required, the number of courses associated with this category.
 - tbl_lookup stores generic lookup information (e.g., mapping status '1' to 'Draft').

Assignment Descriptions

1. Create the pages as below:

HOME PAGE (PUBLIC)



COURSE DETAILS PAGE (PUBLIC)

CourseShare

HomeCategoriesRecent ReviewsLogin

Course Title Goes Here

Categories: HTML Web Frontend

A short description of the course goes here...

Course Content (Markdown)

This is where the Markdown-rendered course content goes.

- Topic 1
- Topic 2
- Topic 3

Student Reviews

John Doe

★★★★☆

Very helpful course, I learned a lot!

Alice Green

★★★★☆

Good content but could use more examples.

Add Your Review

Your Name

Email

Rating

★★★★★

Review

Write your review here...

Submit Review

CATEGORY LIST PAGE (PUBLIC)

CourseShare

Category Cloud

HTML Java css Spring MySQL

Courses in Category: HTML

Introduction to HTML
Learn the basics of HTML, elements, structure, and semantic markup.
[View Course](#)

Advanced HTML5 Features
Dive deeper into HTML5 multimedia tags, APIs, and best practices.
[View Course](#)

[«](#) [<](#) **1** [2](#) [3](#) [>](#) [»](#)

MOST RECENT REVIEWS PAGE (PUBLIC)

CourseShare

Most Recent Reviews

Introduction to HTML
John Doe
★★★★☆
This course helped me understand web basics and structure...

Advanced CSS Techniques
Alice Green
★★★★★
Amazing examples, detailed explanations, highly recommended...

Java Spring Boot 101
Michael Smith
★★★★☆
Great introduction to Spring Boot and REST API development...

LOGIN PAGE (INSTRUCTOR ONLY)

CourseShare

V

Instructor Login

Username

Password

Login

INSTRUCTOR DASHBOARD (PRIVATE)

CourseShare

Dashboard

Manage Courses

Reviews

Logout

Dashboard Overview

24

Total Courses

15

Published Courses

4

Pending Reviews

8

Total Categories

Quick Actions

Create New Course

Manage Courses

Pending Reviews

CREATE/EDIT COURSE PAGE (INSTRUCTOR)

CourseShare

DashboardManage CoursesLogout

Create New Course

Course Title

Short Description

Course Content (Markdown)

Categories (comma-separated)

Status

Draft

Save Draft

Publish

MANAGE COURSES PAGE (INSTRUCTOR)

CourseShare

DashboardManage CoursesLogout

Manage Courses

+ Create New Course

Title	Status	Categories	Actions
Introduction to HTML	Published	HTML, Web	EditDeleteUnpublish Archive
Java Basics	Draft	Java, Programming	EditDeletePublish Archive

<<

<

1

2

3

>

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PENDING/APPROVED REVIEWS PAGE (INSTRUCTOR)

CourseShare

DashboardReviewsLogout

Pending Reviews

Course: Introduction to HTML
Reviewer: John Doe
★★★★☆
Great course, very clear explanations...
[Approve](#) [Delete](#)

Course: Java Basics
Reviewer: Alice Smith
★★★★☆
Useful content but the pace was a bit fast.
[Approve](#) [Delete](#)

Approved Reviews

Course: Spring Boot 101
Reviewer: Mark Lee
★★★★★
Fantastic introduction to Spring Boot!
[Delete](#)

Course: CSS Fundamentals
Reviewer: Emma Brown
★★★★☆
Well structured and informative.
[Delete](#)

-- THE END --